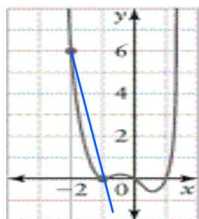


Practice

Part A – Identify the choice that best completes the statement or answers the question.

d 1. A secant drawn through the points shown on the graph has a slope of



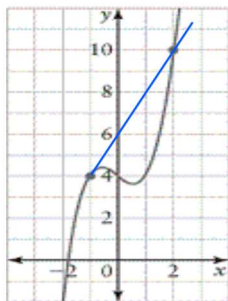
a. -1

c. -3

b. -2

d. -6

b 2. A secant drawn through the points shown on the graph has a slope of



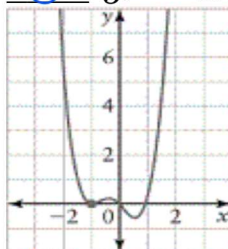
a. $\frac{1}{2}$

c. 5

b. 2

d. $\frac{10}{3}$

C 3. A tangent to the graph of the function at the point shown has a slope of



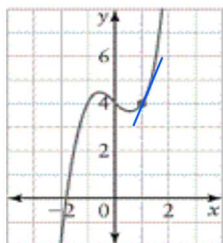
a. -2

c. 0

b. -1

d. 1

b 4. A tangent to the graph of the function at the point shown has a slope of approximately



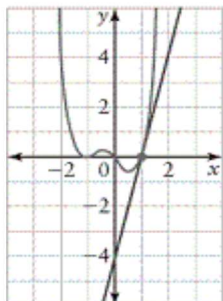
a. $\frac{1}{2}$

c. -2

b. 2

d. 0

d 5. The slope of the tangent at the point indicated on the graph is



a. $\frac{1}{2}$

c. 2

b. 1

d. 4

a

6. The area, A , in square metres, of a circular oil spill after t min is given by $A = kt^2$. The average rate of change of the area of the oil spill over the first 5 min is approximately $15 \text{ m}^2/\text{min}$, the value of k is :

a. 3
b. 6

c. 2
d. 5

$$\frac{A(5) - A(0)}{5} = 15$$

$$25k = 75$$

$$k = 3$$

Part B

1. Suppose the revenue, R , in dollars, from the sales of x (in hundred) MP3 players is given by $R(x) = x(350 - 0.1x^2)$. Find the average rate of change of revenue from selling from 100 to 200 MP3 players.

$$A(1) = 1(350 - 0.1) = 349.9$$

$$A(2) = 2(350 - 0.1(4)) = 699.2$$

$$\text{A.R.O.C.} = \frac{A(2) - A(1)}{2 - 1} = 349.3$$

$\therefore$ The average rate of change is $\$349.30/100$ MP3 players

2. A medical researcher establishes that a patient's reaction time, r , in minutes, to a dose of a particular drug is $r(d) = -0.8d^3 + d^2$, where d is the amount of the drug, in millilitres, that is absorbed into the patient's blood. Determine the instantaneous rate of change of the time reaction with respect to the amount of the drug at 1 millilitre.

$$\text{I.R.O.C.} = \frac{f(1+0.001) - f(1)}{0.001}$$

$$= \frac{0.199599 - 0.2}{0.001}$$

$$\approx -0.4$$

$$f(1.001) = -0.8(1.001)^3 + (1.001)^2$$

$$\approx 0.199599$$

$$f(1) = -0.8 + 1$$

$$= 0.2$$

$\therefore$ The instantaneous rate of change is -0.4 min/mL

3. The following table represents the growth of a bacteria population over a 4.5 h period.

Time (h)	0	0.75	1.5	2.25	3	3.75
Number of bacteria	850	1122	1481	1954	2577	3400

Estimate the instantaneous rate of change in number of bacteria at exactly $t = 3$.

$$\text{I.R.O.C} = \frac{f(3.75) - f(2.25)}{3.75 - 2.25} = \frac{3400 - 1954}{1.5} = 964$$

$\therefore$ The instantaneous rate of change is 964 bacteria/h

4. For the function $f(x) = -2x^3 + 3x - 1$, determine the slope of a tangent at $x = 1$

$$\begin{aligned} \text{I.R.O.C} &= \frac{f(1.001) - f(1)}{0.001} \\ &= \frac{-0.003006}{0.001} \\ &= -3 \end{aligned}$$

$$\begin{aligned} f(1.001) &= -0.003006 \\ f(1) &= 0 \end{aligned}$$

$\therefore$ The slope is -3

5. Find the equation of tangent line to the curve $f(x) = -x^4 + 1$ at $x = -1$ on the curve.

$$\begin{aligned} \text{I.R.O.C} &= \frac{f(-1+0.001) - f(-1)}{0.001} \\ &= \frac{0.003994}{0.001} \\ &= 4 \end{aligned}$$

$$\begin{aligned} f(-0.999) &= 0.003994 \\ f(-1) &= -(-1)^4 + 1 = 0 \end{aligned}$$

$$\text{Sub. } (-1, 0) \text{ into } y = 4x + b: \begin{aligned} 0 &= -4 + b \\ 4 &= b \end{aligned}$$

$$\therefore y = 4x + 4$$